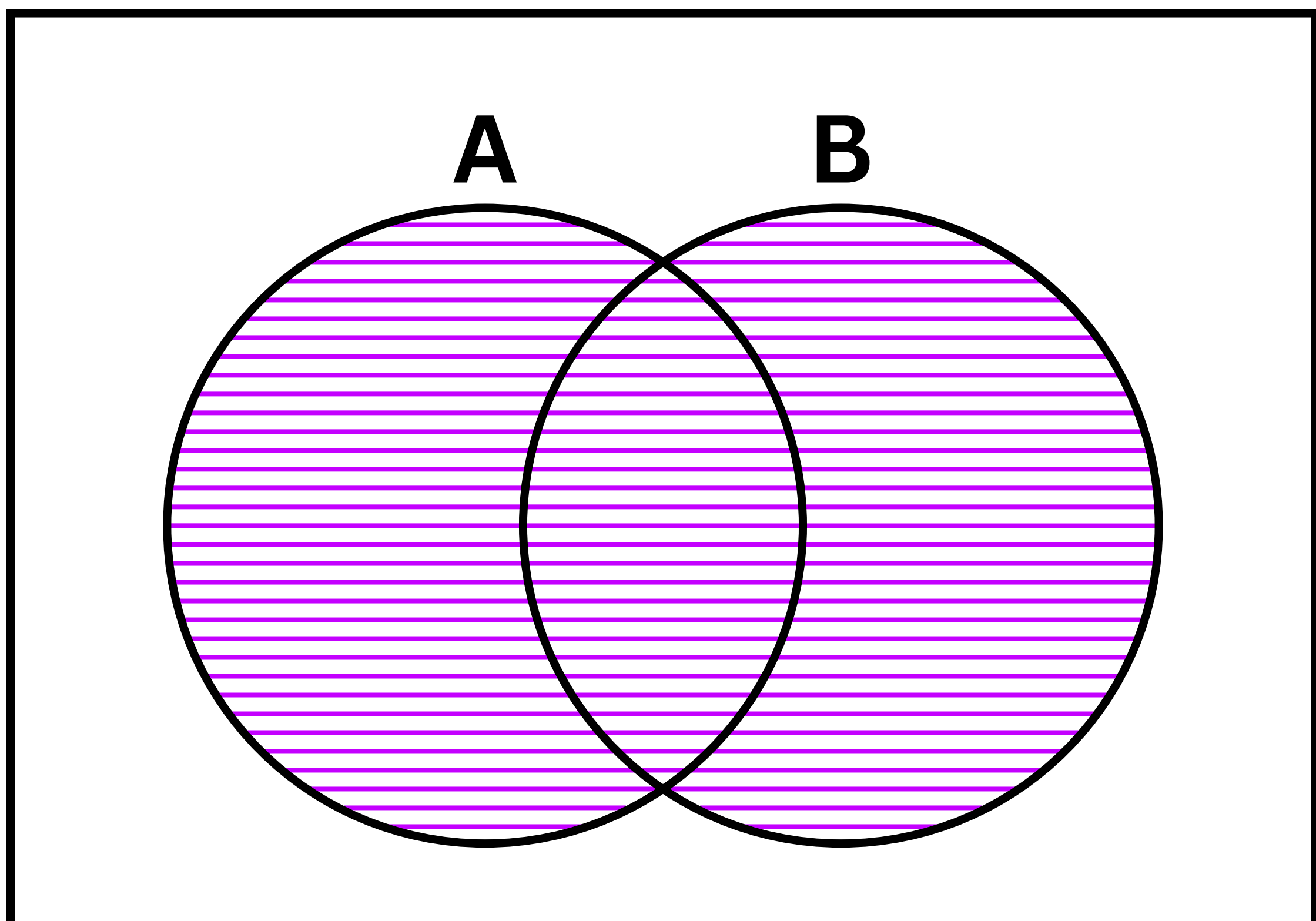


Probability

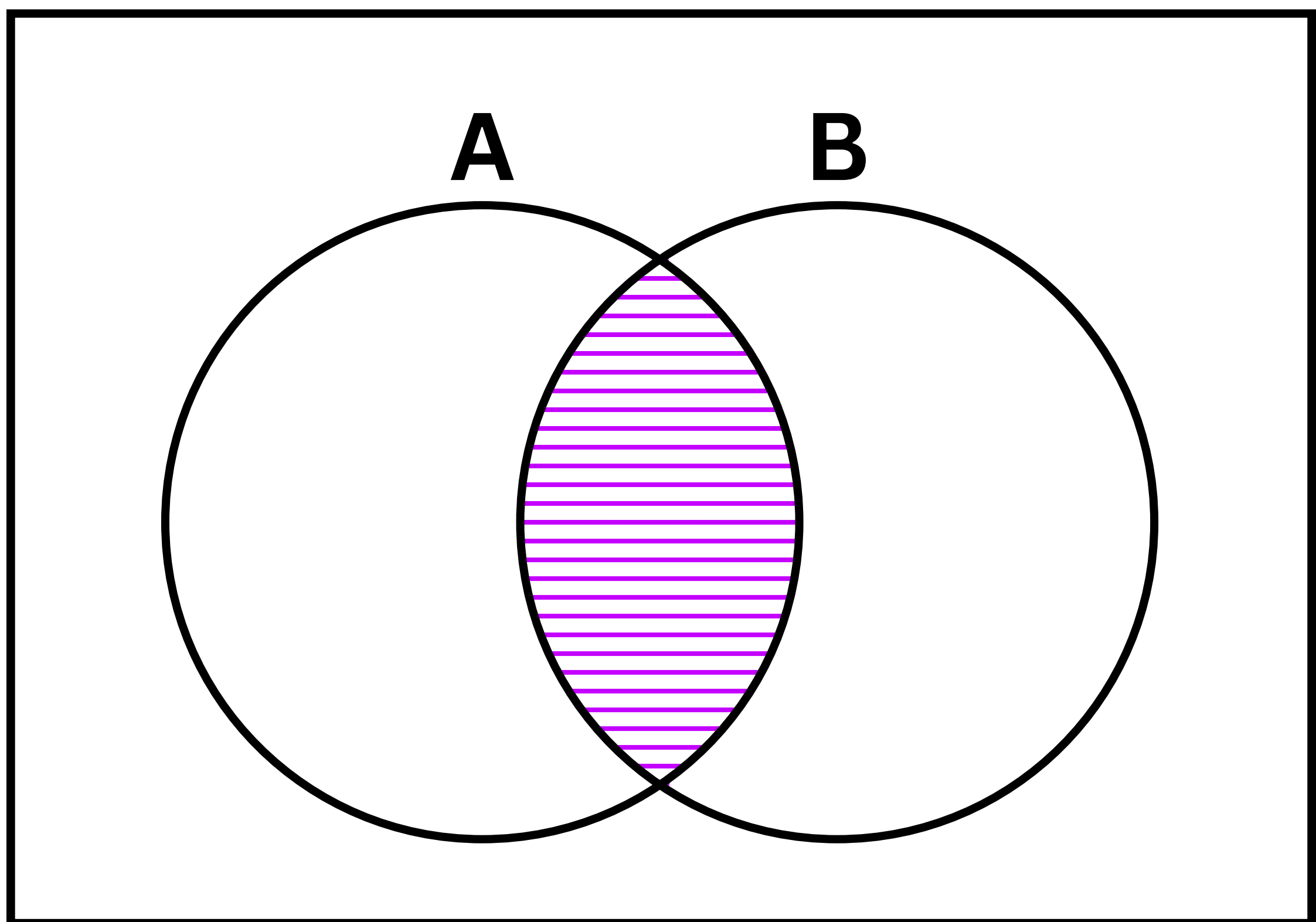
$$A \text{ or } B \rightarrow A \cup B$$

‘Event A or B’ refers to the occurrence of either A or B. (At least A or B).



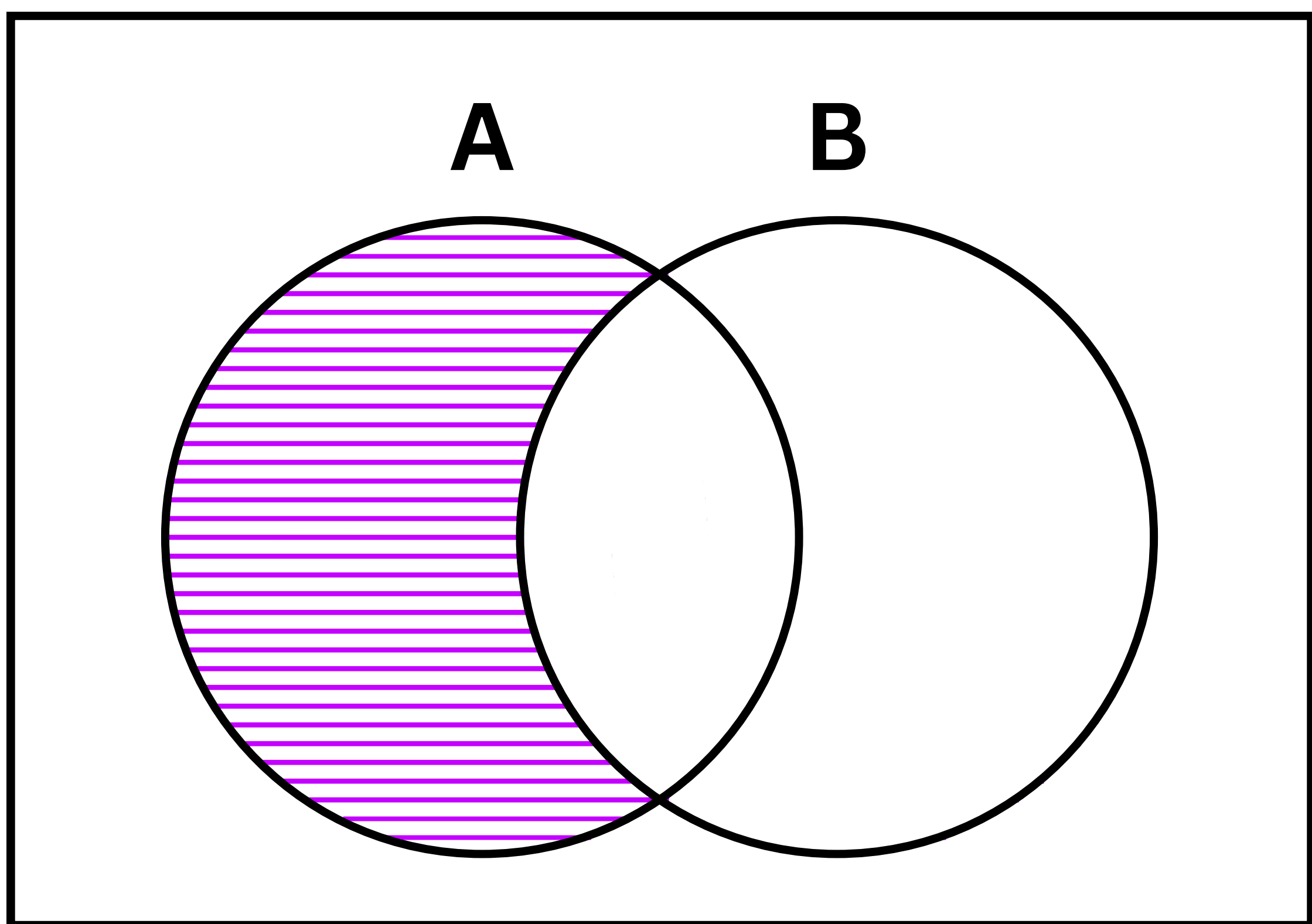
$$A \text{ and } B \rightarrow A \cap B$$

‘Event A and B’ refers to the occurrence of A and B simultaneously.



$$A \text{ but not } B \rightarrow A \cap B'$$

“A but not B” refers to the occurrence of only event A and not event B.



Classical Definition of Probability

$$P(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

$$P(A) + P(\bar{A}) = 1$$

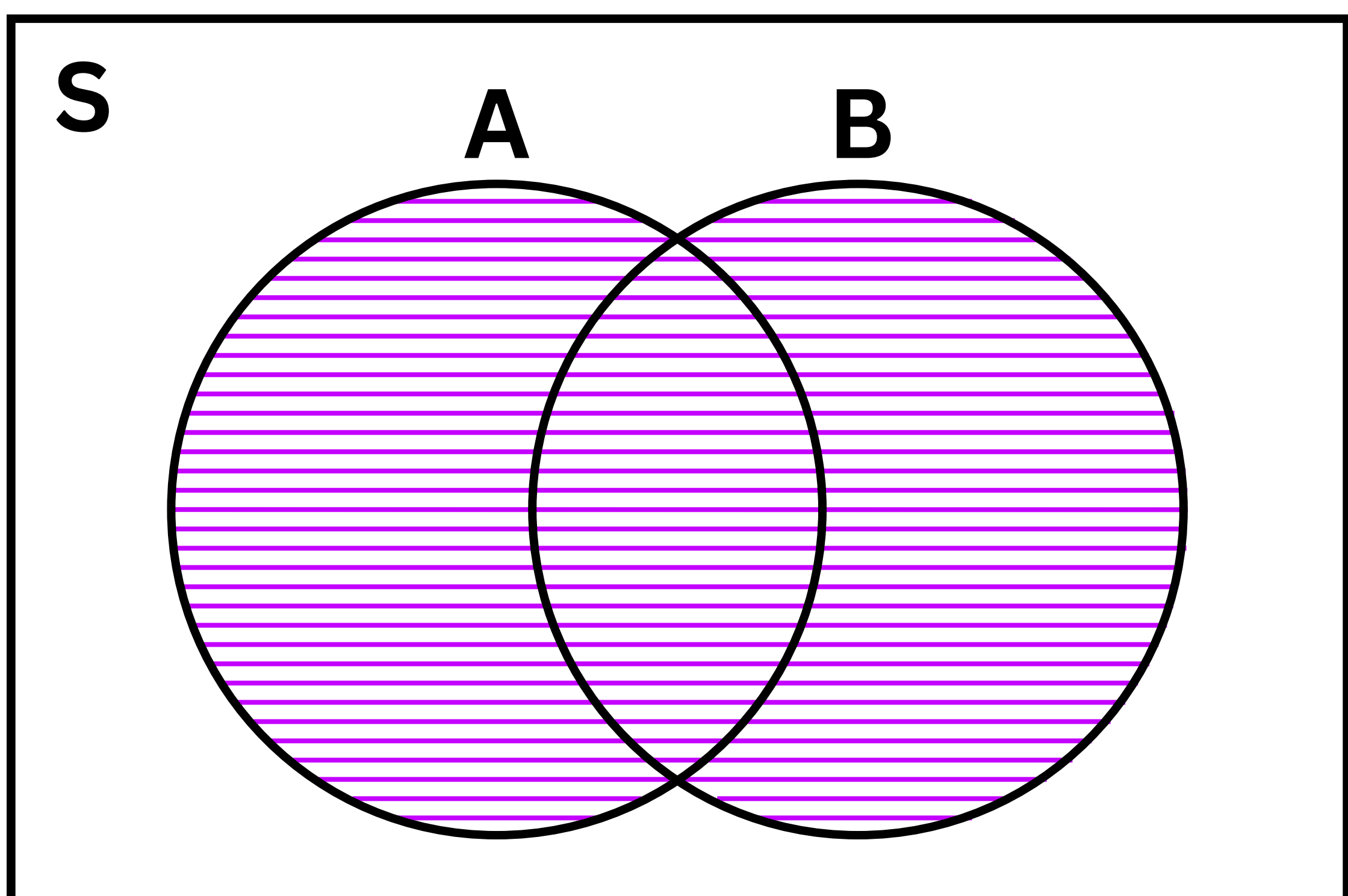
$$0 \leq P(A) \leq 1$$

Addition Theorem of Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

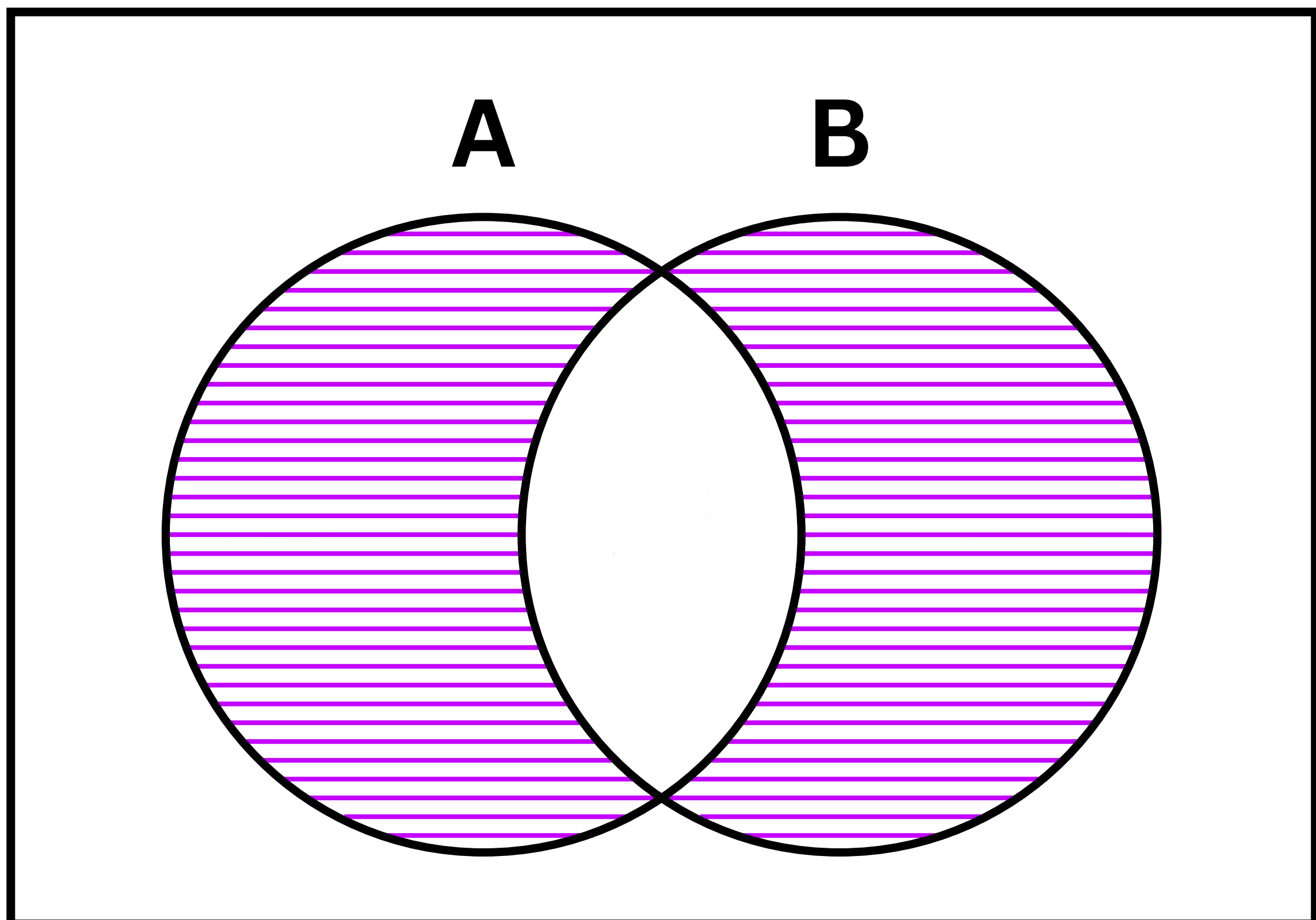
$$\frac{n(A \cup B)}{n(S)} = \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



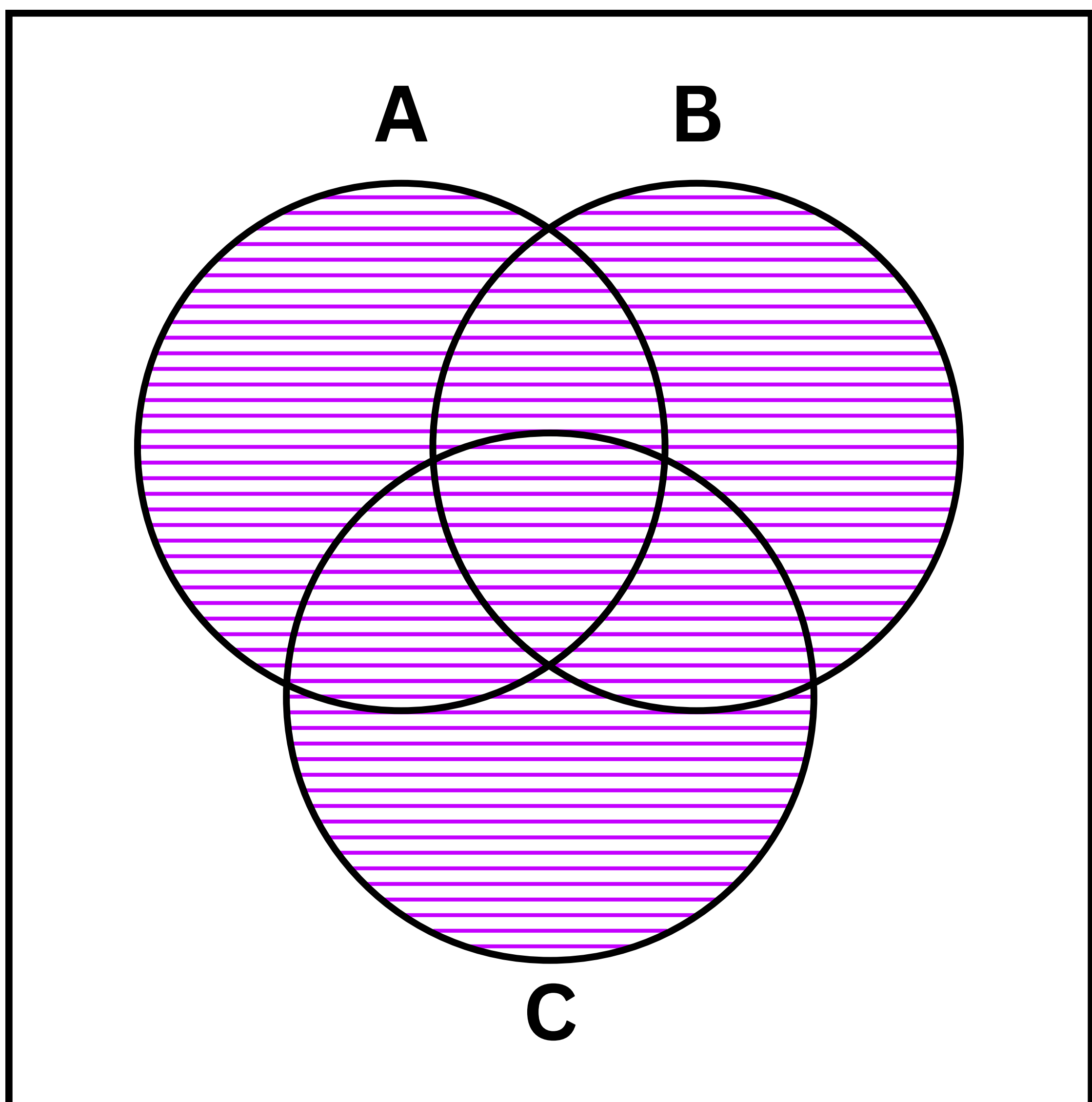
Probability of Exactly A or Exactly B Happening:

$$P \left[(\bar{A} \cap B) \cup (A \cap \bar{B}) \right] \\ = P(A) + P(B) - 2P(A \cap B)$$



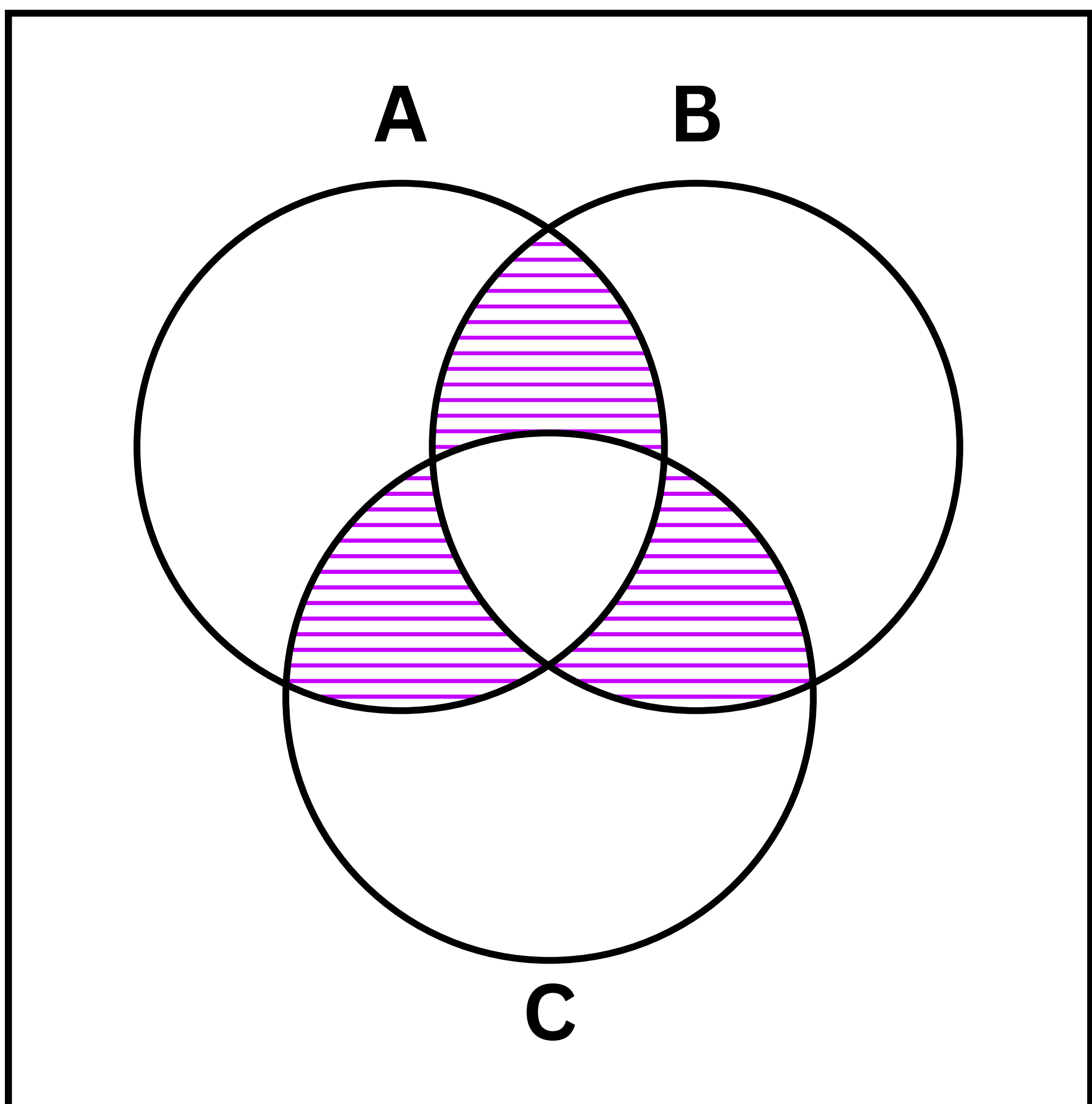
Probability of at least one of the three events A, B or C happening:

$$\begin{aligned} &P(A \cup B \cup C) \\ &= P(A) + P(B) + P(C) \\ &- P(A \cap B) - P(B \cap C) - P(A \cap C) \\ &+ P(A \cap B \cap C) \end{aligned}$$



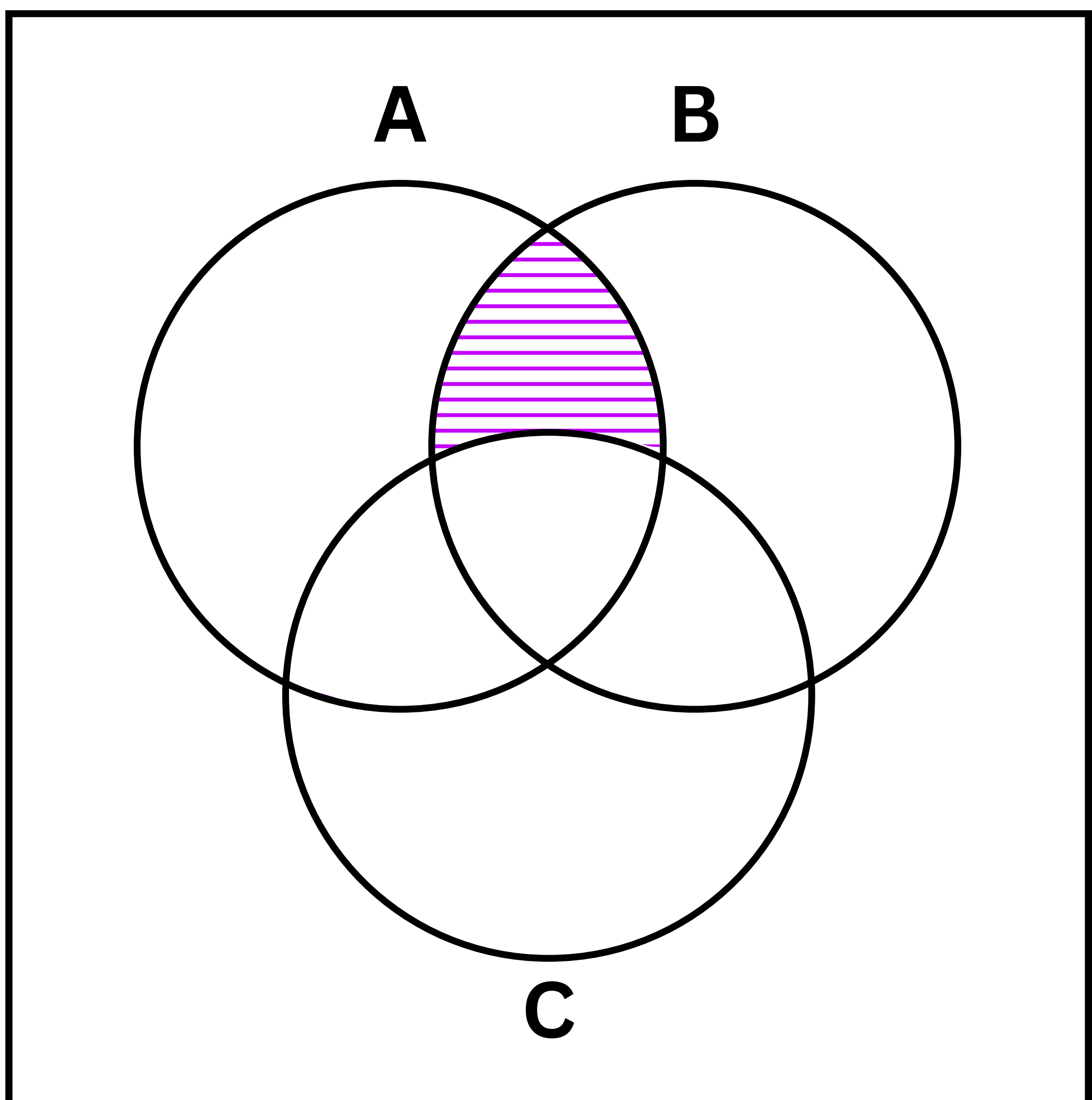
Probability of exactly two of the three events, A, B, or C, occurring simultaneously:

$$= P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$$



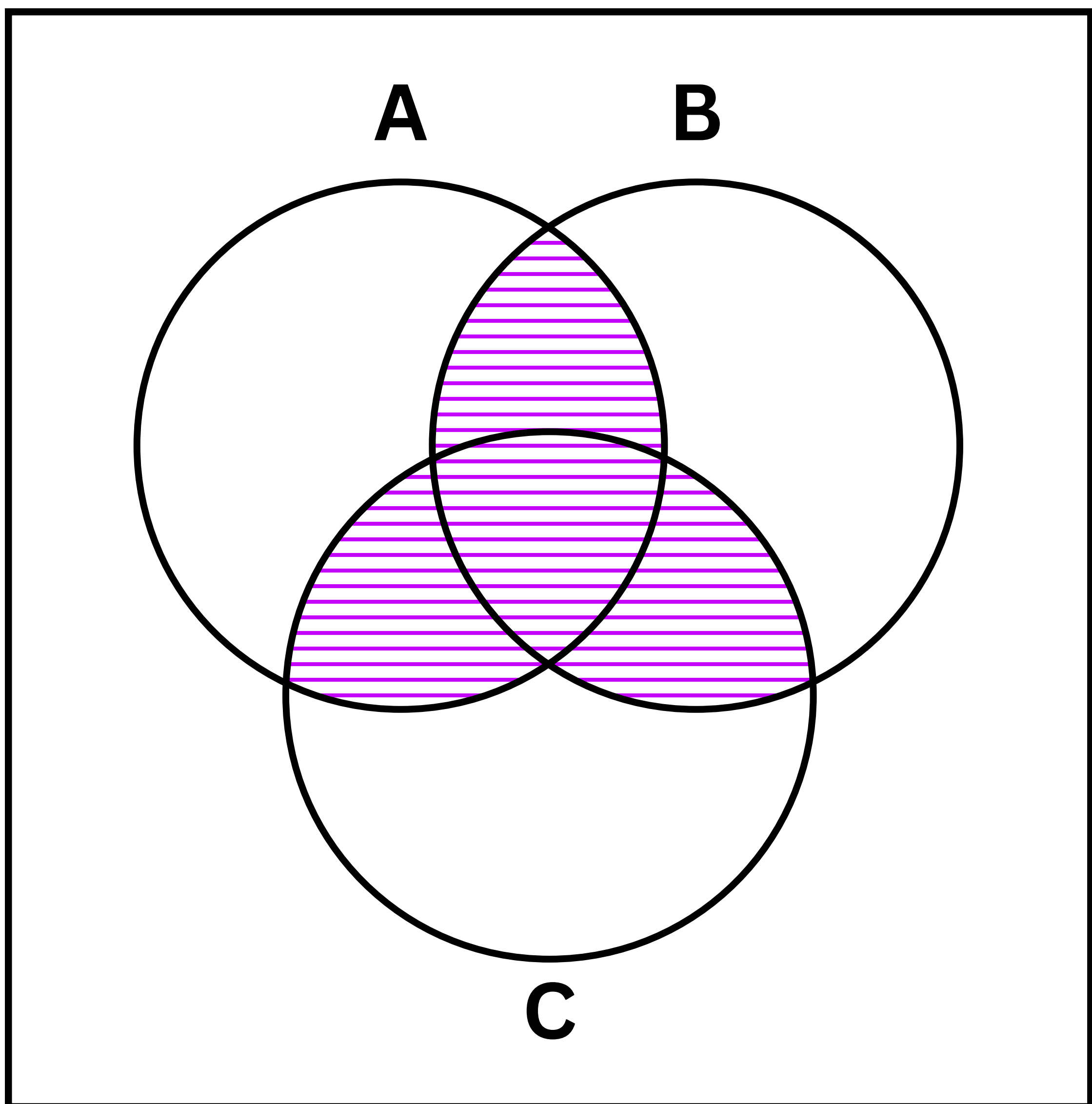
Probability of only two specific events happening simultaneously (A and B), while a third event (C) does not occur:

$$\begin{aligned} & (P(A \cap B \cap \bar{C})) \\ &= P(A \cap B) - P(A \cap B \cap C) \end{aligned}$$



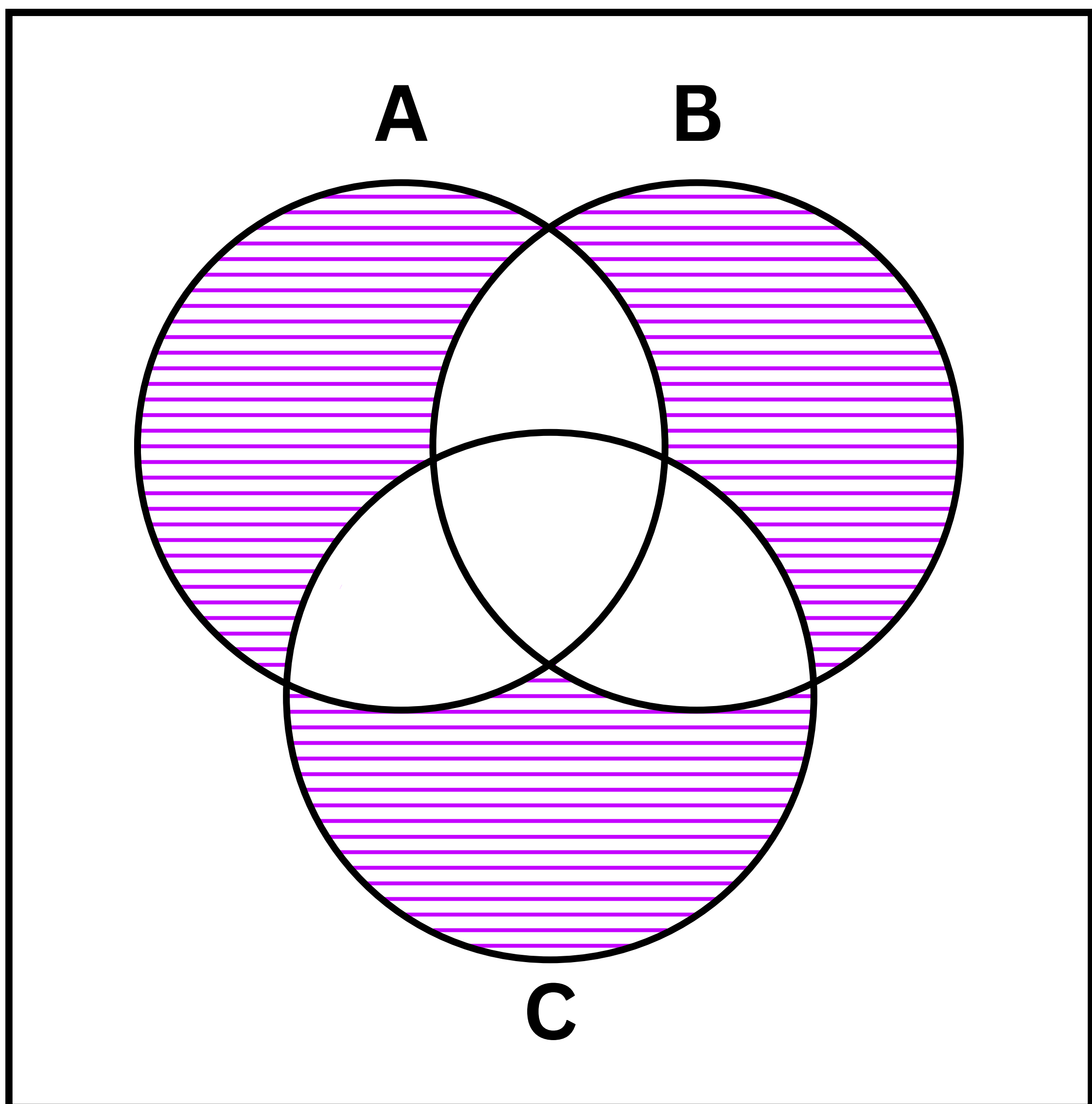
**Probability of at least two of
the three events A , B , or C
occurring at the same time:**

$$\begin{aligned} &P(\text{at least two of } A, B, C) \\ &= P(A \cap B) + P(A \cap C) + P(B \cap C) \\ &\quad - 2P(A \cap B \cap C) \end{aligned}$$



Probability of exactly one of three events (A, B, or C) occurring:

$$\begin{aligned} &P(\text{exactly one of } A, B, C) \\ &= P(A) + P(B) + P(C) \\ &- 2P(A \cap B) - 2P(A \cap C) - 2P(B \cap C) + \\ &\quad 3P(A \cap B \cap C) \end{aligned}$$



Conditional Property:

Probability of Event A when Event B has already occurred →

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of Event B when Event A has already occurred →

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

NOTE →

$$P(A|B) + P(\bar{A}|B) = 1$$

If A and B are M.E.E. (Mutually Exclusive Events), then →

$$P(A|B) = P(B|A) = 0$$

$$(\because A \cap B = \phi)$$

Mutually exclusive events cannot happen together (making them dependent), while independent events do not affect each other's probability, meaning they can often happen together. In short, mutually exclusive means "no overlap," while independence means "no influence."

Multiplication Rule for Independent Events

Conditional Probability (General) →

The general formula calculates the probability of event A happening given that event B has already happened, effectively restricting the sample space to only include outcomes where B occurs.

If A and B are truly independent events, knowing that event B has occurred provides no new information about event A. Therefore, the conditional probability of A given B is simply equal to the original (unconditional) probability of A:

When A and B are Independent Events $\rightarrow$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

$$P(A \cap B) = P(B) \cdot P(A|B) = P(B) \cdot P(A)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

(A and B are Independent Events)

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

(A, B, C are Independent Events)

When A and B are Independent Events,
the following is true as well :

$$P(A' \cap B) = P(A') \cdot P(B)$$

$$P(A \cap B') = P(A) \cdot P(B')$$

$$P(A' \cap B') = P(A') \cdot P(B')$$

De Morgan's Law :

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

Therefore :

$$P(\overline{A \cup B \cup C}) = 1 - P(A \cup B \cup C)$$

$$P(A \cup B \cup C) = 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

Multiplication Rule:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

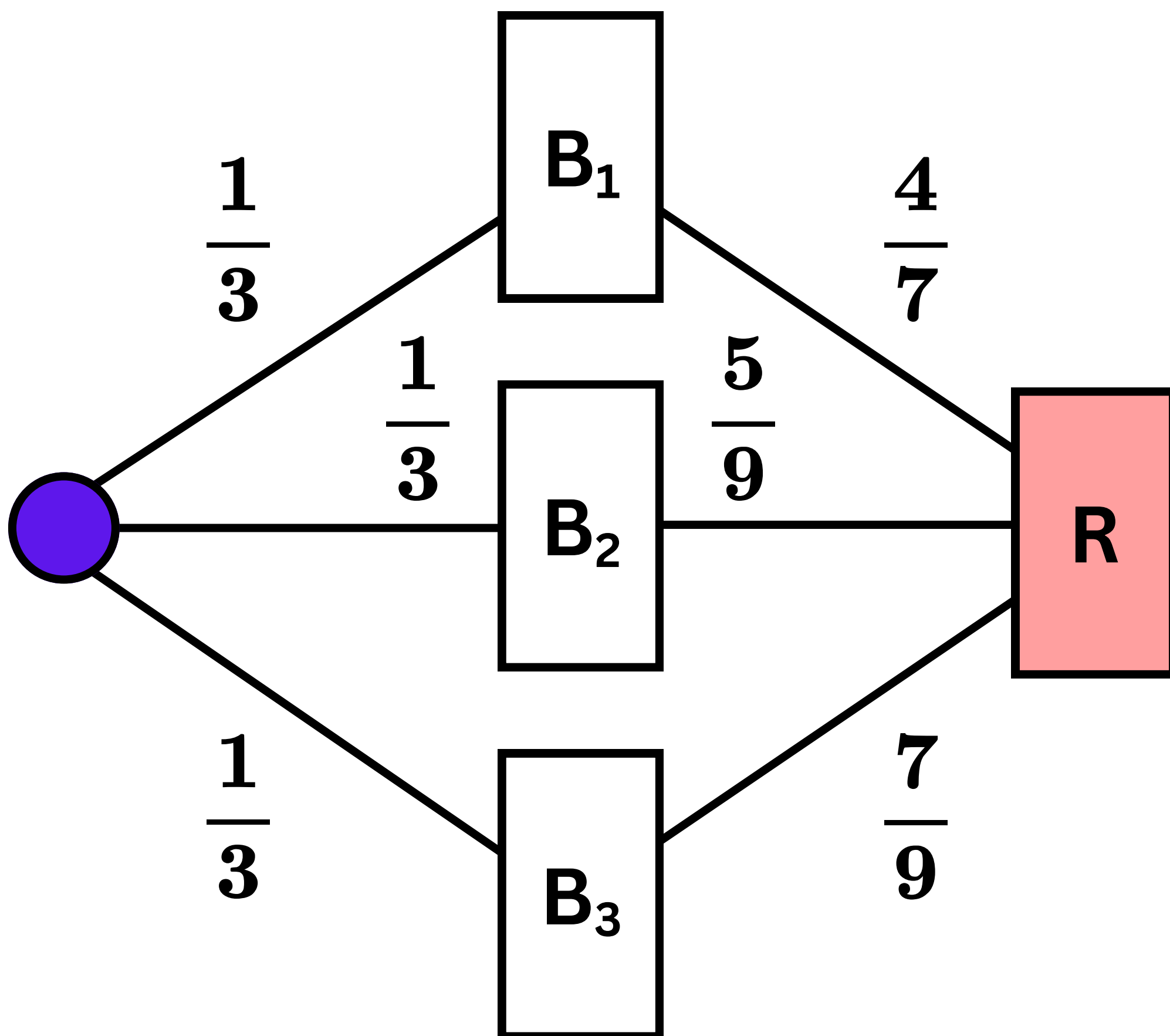
$$P(A \cap B) = P(B) \cdot P(A|B)$$

Total Probability Theorem: (Learning via Application)

Q: A bag is selected at random from the three bags B_1 , B_2 , and B_3 . A single ball is then drawn at random from the selected bag. Find the probability that the ball drawn is red.

B_1	B_2	B_3
3 W	4 W	2 W
4 R	5 R	7 R

A: Draw a 'probability tree' and write the probability of each outcome above its respective 'branch' as shown (in the next page).



Probability of Getting a Red Ball

$$= \left(\frac{1}{3} \times \frac{4}{7} \right) + \left(\frac{1}{3} \times \frac{5}{9} \right) + \left(\frac{1}{3} \times \frac{7}{9} \right)$$

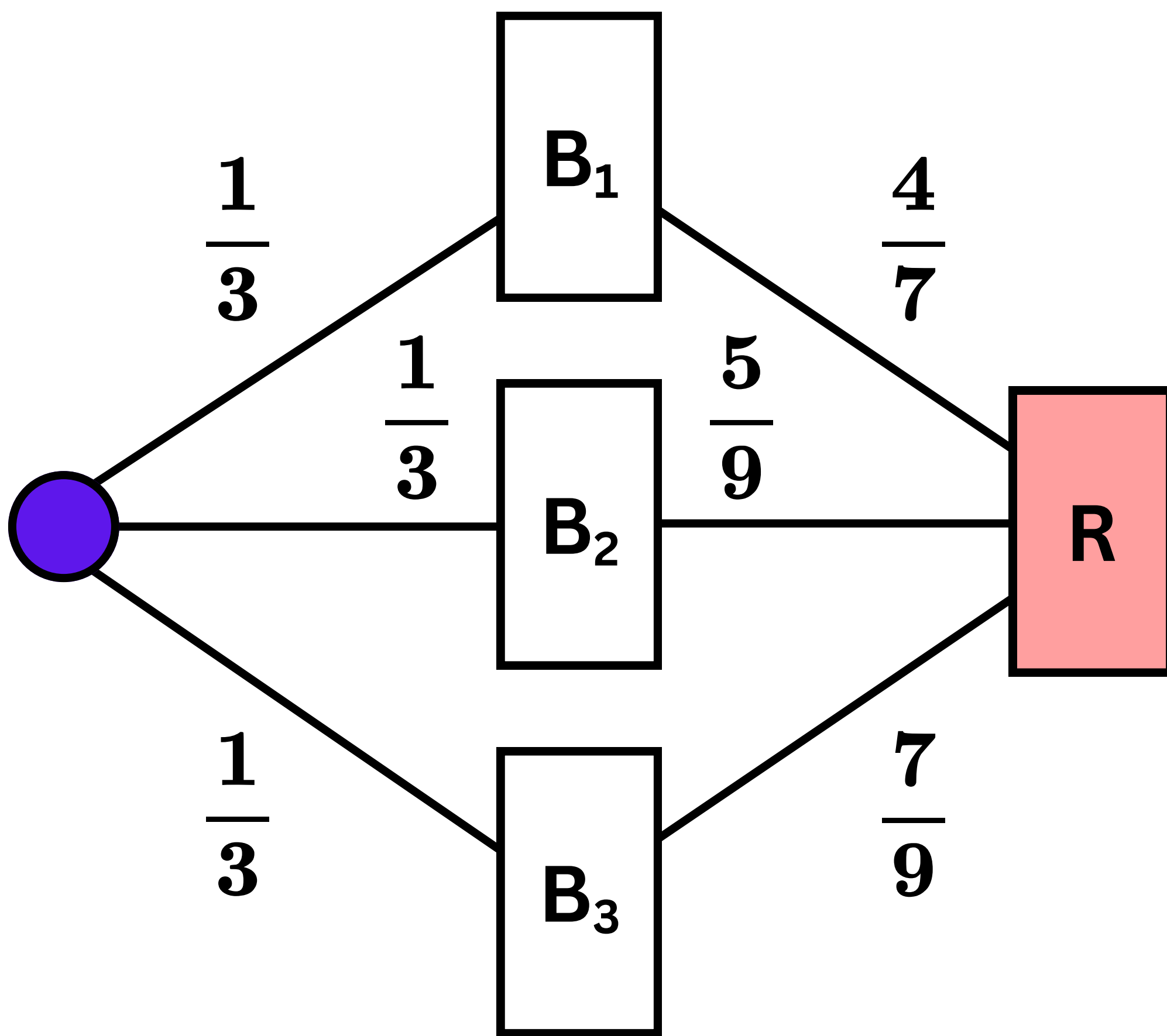
Inverse Probability Theorem:
(Bayes' Theorem)
(Learning via Application)

Over here, we divide the probability of a single complete 'branch' (path) by the total probability of the 'tree'.

- The numerator is the probability of the specific "path" (branch) that leads to the outcome you observed.
- The denominator is the sum of the probabilities of all possible paths (all relevant branches across the whole tree) that could result in that same outcome.

Q: A bag is selected at random from the three bags B_1 , B_2 , and B_3 , and a ball is drawn. Given that the ball drawn is red, what is the probability that the selected bag was Bag 2?"

B_1	B_2	B_3
3 W	4 W	2 W
4 R	5 R	7 R



Probability of a Red Ball From B_2

$$= \frac{\left(\frac{1}{3} \times \frac{5}{9}\right)}{\left(\frac{1}{3} \times \frac{4}{7}\right) + \left(\frac{1}{3} \times \frac{5}{9}\right) + \left(\frac{1}{3} \times \frac{7}{9}\right)}$$

Binomial Probability Distribution:

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

$$P(0) + P(1) + P(2) + \dots + P(n) = 1$$

$$(p + q = 1)$$

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

- **n = number of trials.**
- **X = r = number of successes desired in n trials.**
- **p = probability of success in a single trial.**
- **q = probability of failure in a single trial (p + q = 1).**

Random Variable

A random variable is simply a variable whose value depends on the outcome of a random event. Let us use the example of tossing two coins simultaneously:

- The possible outcomes are {HH, HT, TH, TT}.
- If we define 'X' as the number of heads obtained, X is the random variable.
- X can take the values 0 (for TT), 1 (for HT or TH), or 2 (for HH).

Probability Distribution

A probability distribution is a function or table that lists all the possible values a random variable can take and their associated probabilities. It tells you how likely each outcome is.

- For the coin toss example, this would be a table showing:
 - Probability of **0 heads (TT):**
 $\frac{1}{4}$ or 0.25
 - Probability of **1 head (HT, TH):**
 $\frac{2}{4}$ or 0.50
 - Probability of **2 heads (HH):**
 $\frac{1}{4}$ or 0.25

Q: A pair of fair dice is thrown three times. If getting a sum of 9 is considered a success, what is the probability distribution for the number of successes?

X	0	1	2	3
P(X)	P₀	P₁	P₂	P₃

No. of trials = $n = 3$

Successful Outcomes

= (3, 6) , (4, 5) , (5, 4) , (6, 3)

Probability of a Successful Outcome

$$= p = \frac{4}{36} = \frac{1}{9}$$

Probability of Failure

$$= q = 1 - p = \frac{8}{9}$$

$$\mathbf{P}(\mathbf{X} = \mathbf{r}) = {}^{\mathbf{n}}\mathbf{C}_{\mathbf{r}} \mathbf{p}^{\mathbf{r}} \mathbf{q}^{\mathbf{n}-\mathbf{r}}$$

$$\mathbf{P}(\mathbf{X} = \mathbf{0}) = {}^{\mathbf{3}}\mathbf{C}_{\mathbf{0}} \left(\frac{\mathbf{1}}{\mathbf{9}}\right)^{\mathbf{0}} \left(\frac{\mathbf{8}}{\mathbf{9}}\right)^{\mathbf{3}} = \mathbf{P}_{\mathbf{0}}$$

$$\mathbf{P}(\mathbf{X} = \mathbf{1}) = {}^{\mathbf{3}}\mathbf{C}_{\mathbf{1}} \left(\frac{\mathbf{1}}{\mathbf{9}}\right)^{\mathbf{1}} \left(\frac{\mathbf{8}}{\mathbf{9}}\right)^{\mathbf{2}} = \mathbf{P}_{\mathbf{1}}$$

$$\mathbf{P}(\mathbf{X} = \mathbf{2}) = {}^{\mathbf{3}}\mathbf{C}_{\mathbf{2}} \left(\frac{\mathbf{1}}{\mathbf{9}}\right)^{\mathbf{2}} \left(\frac{\mathbf{8}}{\mathbf{9}}\right)^{\mathbf{1}} = \mathbf{P}_{\mathbf{2}}$$

$$\mathbf{P}(\mathbf{X} = \mathbf{3}) = {}^{\mathbf{3}}\mathbf{C}_{\mathbf{3}} \left(\frac{\mathbf{1}}{\mathbf{9}}\right)^{\mathbf{3}} \left(\frac{\mathbf{8}}{\mathbf{9}}\right)^{\mathbf{0}} = \mathbf{P}_{\mathbf{3}}$$

Mean and Variance: Random Variable

Mean ($\mu, \bar{x}$) :

This is the average or central value of a random variable, also known as the Expectation: $E(X)$.

In simple terms: If you repeated a random event many times (like rolling a die), the mean is the long-term average outcome you would expect.

Formula :
$$\frac{\sum P_i X_i}{\sum P_i} = \sum P_i X_i$$

This calculates a weighted average. You multiply each possible outcome X_i by its probability of happening P_i and then add all those products together. The bottom part of the fraction, $\sum P_i$ equals 1 for a full probability distribution, so it's usually just written as $\sum P_i X_i$.

Variance (σ^2) :

In simple terms: A low variance means the outcomes tend to cluster closely around the average, while a high variance means they are widely dispersed.

Formula : $\sigma^2 = \Sigma(X_i - \mu)^2 \cdot P_i$

$$\sigma^2 = \Sigma(P_i X_i^2) - \mu^2$$

Standard Deviation (S.D.) :

Formula : $\text{S.D.} = \sqrt{\text{Variance}}$

X	2	3	4
P(X = x)	0.3	0.4	0.3

Mean :

$$\mu = \Sigma X_i P_i$$

$$\mu = (2 \times 0.3) + (3 \times 0.4) + (4 \times 0.3)$$

$$\mu = 0.6 + 1.2 + 1.2$$

$$\mu = 3.0$$

Variance :

$$\sigma^2 = \Sigma (X_i - \mu)^2 P_i$$

$$\sigma^2 = (2 - 3)^2 \times 0.3$$

$$+ (3 - 3)^2 \times 0.4$$

$$+ (4 - 3)^2 \times 0.3$$

$$\sigma^2 = (-1)^2 \times 0.3$$

$$+ (0)^2 \times 0.4$$

$$+ (1)^2 \times 0.3$$

$$\sigma^2 = 1 \times 0.3 + 0 \times 0.4 + 1 \times 0.3$$

$$\sigma^2 = 0.3 + 0 + 0.3$$

$$\sigma^2 = 0.6$$

Standard Deviation (S. D.) :

$$\sigma = \sqrt{\text{Variance}}$$

$$\sigma = \sqrt{0.6}$$

Mean and Variance: Binomial Variate

Mean ($\mu, \bar{x}$) :

Formula : $\mu = np$

(number of trials $\times$ probability of success)

Variance (σ^2) :

Formula : Variance = npq

(mean $\times$ probability of failure)

Standard Deviation : S. D.

Formula : S.D. = $\sqrt{\text{variance}}$

**If X follows a binomial distribution,
with mean = 3 and variance = $\frac{3}{2}$, find :**

(i) $P(X \geq 1)$ (ii) $P(X \leq 5)$

$$\text{Variance} = npq = \frac{3}{2}$$

$$\text{Mean} = np = 3$$

$$q = p = \frac{1}{2}$$

$$n \times \frac{1}{2} = 3 \rightarrow n = 6$$

(i) $P(X \geq 1)$

$$= 1 - P(X = 0)$$

$$= 1 - {}^6C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^6$$

(ii) $P(X \leq 5)$

$$= 1 - P(X = 6)$$

$$= 1 - {}^6C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^0$$